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Task 2 task report:

1. Problem

A smart classroom agent reads temperature and occupancy, then picks a mode: ECO, COOL, WARM or IDLE. An empty room always has priority. At exactly 20°C or 26°C the mode is IDLE because the code uses < and >.

2. PEAS

	Delivery Robot	LLM Student-Support Agent
Performance	On-time delivery, no collisions, battery use	Correct answers, student satisfaction, privacy
Environment	Campus paths, pedestrians, weather	Students, university FAQ, portal
Actuators	Wheels, brake, horn, parcel lock	Reply, FAQ search, create ticket, staff handoff
Sensors	Camera, lidar, GPS, battery level	Student message, chat history, documents
Trade-off	Speed vs safety	Fast reply vs unsupported answer

The helpdesk agent can only use read-only FAQ search and ticket creation. It has no access to grades or payments.

3. Environment Classification

Both the delivery robot and the helpdesk agent are partially observable and stochastic. The classroom simulator is fully observable and deterministic, but only under our simplified assumptions.

4. Reflex vs Stateful Agent

The reflex agent has no memory, so it repeats COOL. The stateful agent stores the previous mode and sends a command only when the mode changes. Both choose the same target.

5. Edge Case

Noisy readings around 26°C (26.4, 25.8, 26.3, 25.9) give COOL, IDLE, COOL, IDLE, with a command sent every step. The cause is that remembering the last mode is not hysteresis. The fix is to use two thresholds, such as cool above 26.5 and stop below 25.5.

6. Limitations

Both agents use fixed rules, do not learn, and fully trust the sensors. An LLM agent is more flexible but less predictable, so it needs guardrails.

7. Reflection

PEAS describes the task, not the policy. Memory alone did not stop the flip-flopping. Next time I will add hysteresis first. Performance measures must include safety and privacy, not only speed.